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EXAMINER
ALHARBI, ADAM MOHAMED

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3663	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/123,198	Applicant(s) Fang et al.	
	Examiner ADAM ALHARBI	Art Unit 3663	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 05/12/2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-22 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-22 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 2/8/2025 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of Claims

This Office Action is in response to the application filed on May 12, 2025. Claim 1 has been amended. Claims 1-22 are presently pending and are presented for examination.

Response to Amendments

In response to the Amendments dated May 12, 2025, the Examiner withdraws the previous art rejections.

Response to Arguments

Applicant's arguments filed May 12, 2025 have been fully considered but they are moot in view of the new ground(s) of rejection.

Information Disclosure Statement

The Applicant has submitted an information disclosure statement (IDS) which is 38 pages long that includes 32 *pages* of US Patent or Pre-Grant Publications followed by a list of 18 Foreign references and 23 Non-Patent Literature (NPL) references, likely totaling thousands upon thousands of pages to potentially review. In accord with dicta from Molins PLC v. Textron, Inc., 48 F.3d 1172 (Fed. Cir. 1995), stating that forcing the Examiner to find "a needle in a haystack" is "probative of bad faith." Id. [The

Molins] case presented a situation where the disclosure was in excess of 700 pages and contained more than fifty references. Likewise, the instant application's IDS includes way more than even what was found to be excessive in the Molins case, and this IDS does not include any concise explanation of the relevance of any of the listed references nor cite any pages, columns, and lines (or paragraph numbers) where relevant passages or relevant figures appear. According to MPEP Section 2004 "Aids to Compliance With Duty of Disclosure [R-08.2012]", "It is desirable to *avoid the submission of long lists of documents* if it can be avoided. *Eliminate clearly irrelevant and marginally pertinent cumulative information. If a long list is submitted, highlight those documents which have been specifically brought to Applicant's attention and/or are known to be of most significance*." Additionally, per MPEP Section 609.04(a)(III): "*applicants are encouraged to provide a concise explanation of why the English-language information is being submitted and how it is understood to be relevant. Concise explanations (especially those which point out the relevant pages and lines) are helpful to the Office*, particularly where documents are lengthy and complex and applicant is aware of a section that is highly relevant to patentability or *where a large number of documents are submitted and applicant is aware that one or more are highly relevant to patentability*." See *Penn Yan Boats, Inc. v. Sea Lark Boats, Inc.*, 359 F. Supp. 948, 175 USPQ 260 (S.D. Fla. 1972), *aff'd,* 479 F.2d 1338, 178 USPQ 577 (5th Cir. 1973), *cert. denied,* 414 U.S. 874 (1974). But cf. *Molins PLC v.Textron Inc.*, 48 F.3d 1172, 33 USPQ2d 1823 (Fed. Cir. 1995).

As such, even though this IDS has been placed in the application file with the list of references marked as *considered,* and the compilation of those listed US Patents and Pre-Grant Publications have at least been key-word searched and/or classification searched for relevant prior art, and the compilation of foreign and/or NPL references have at least been briefly reviewed (for example, abstracts and drawings reviewed), the information referred to therein for each individual reference has admittedly been considered only within the reasonable time period allocated by the Examiner out of the total normal examination time given for any given application with a similar classification picture. If Applicant

wishes to have one or more references considered more in depth, the Examiner requests resubmitting the IDS with a *reasonable number of references that are known to be pertinent for the determination of patentability as defined by 37 C.F.R. § 1.56*, along with the concise explanations as to relevance and citations explaining the locations of relevant passages or figures, as per 37 CFR 1.98(a)(3) and 37 CFR § 1.105.

Double Patenting

Claim 1 is provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1 and 24 of parent Application No. 18/123172 as follows:

Current Application's claims	Parent Application's claims
1	1 and 24

Although the claims at issue are not identical, they are not patentably distinct from each other because current claims additionally include "interpret a plurality of vehicle parameter values from multiple endpoints". Reference in 103 rejection below teaches the limitation and therefore it is obvious to modify the parent application to include the limitation(s) in this application.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference

claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq*. for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to ATA 35 U.S.C. 102 and 103 is incorrect, any correction of the statutory basis for the rejection will not be considered a new

ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims 1-4 and 6 are rejected under 35 U.S.C. 103 as being unpatentable over U.S. Pub. No. 2011/0130916 (hereinafter, "Mayer"; previously of record) in view of U.S. Pub. No. 2016/0171793 (hereinafter, "Kim"; newly of record).

Regarding claim **1**, Mayer discloses **an apparatus structured to be disposed onboard a vehicle, the apparatus comprising:**

a parameter acquisition circuit structured to interpret a plurality of vehicle parameter values ("a data processing module 34 that processes the received data" (para 0114))**, responsive to the at least one requested vehicle property** ("the RDS server 30 then filters and reports the requested items in modified messages M2* to each respective user which each contain only the information items requested by that user" (para 0097))**, from a plurality of providing end points** ("collecting raw data from the various vehicle sensors and operational components" (para 0076))**, each of the plurality of providing end points on at least one network zone of a vehicle** (" The parameters needed to determine such a condition would stem from vehicle components that are nodes (or message sources) on the vehicle network(s) that the RDU 20 is connected to via CAN bus 18" (para 0119))**; and**

a vehicle data transmission circuit structured to selectively transmit at least a portion of a collected vehicle data ("active vehicles transmit a partial data status message M2 containing selected CAN and GPS information items to the RDS" (para 0126))**;**

However, Mayer does not explicitly teach

a policy acquisition circuit structured to interpret a vehicle policy data value comprising at least one requested vehicle property; and

an apparatus structured to be disposed onboard a vehicle.

Kim, in the same field of endeavor, teaches

a policy acquisition circuit structured to interpret a vehicle policy data value comprising at least one requested vehicle property ("an apparatus for processing a plurality of logging policies and a method thereof, and more particularly, to techniques for processing a plurality of logging policies without having mutual collision in collecting vehicle data (e.g., sensor data, control data, network data,

etc.)" (para 0002) and "a logging policy input section configured to receive a plurality of logging policies for use with vehicle data" (para 0007)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Kim in order to selectively collect the vehicle data based on a logging policy; see Kim at least at [0004]**; and**

an apparatus structured to be disposed onboard a vehicle ("an apparatus for processing a plurality of logging policies according to embodiments of the present disclosure is mounted to the vehicles to process a plurality of logging policies applied to vehicles" (para 0038) and "A logging policy of the plurality of logging policies may include one or more policy files in a script format, the profile data, the variable data, which is used to determine a time of applying the policy, and the policy data, which is used to collect the vehicle data" (para 0020)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Kim in order to selectively collect the vehicle data based on a logging policy; see Kim at least at [0004].

Regarding claim **2**, Mayer discloses and Kim teaches **the apparatus of claim 1**. Additionally, Mayer discloses **wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data by selecting a transmission interval for the at least a portion of the collected vehicle data** ("the partial data status message need only be sent at time intervals T2 " (para 0126))**.**

Regarding claim **3**, Mayer discloses and Kim teaches **the apparatus of claim 2**. Additionally, Mayer discloses **wherein the vehicle data transmission circuit is further structured to select the transmission interval in response to at least one of:**

an interval provided in the vehicle policy data value ("creates an event window of a predetermined time interval about the time stamp, and selects a portion of the logged data falling within the predetermined time interval" (para 0017));

an interval responsive to an availability description for transmitting resources for the at least a portion of the collected vehicle data ("During these intervals, in addition to any communications associated with keeping the wireless link up, the processor 24 may send additional real time vehicle data" (para 0082));

an interval responsive to a historical transmission availability for the vehicle; or

an operating condition of the vehicle ("transmit one or more predetermined status messages or vehicle data to the backend or server side of the system at predetermined intervals" (para 0082) and "the data logging system includes a communication module which communicates with engine and other vehicle components over a communication channel or bus to receive data from the vehicle components" (para 0016)).

Regarding claim **4**, Mayer discloses and Kim teaches **the apparatus of claim 2**. Additionally, Mayer discloses **wherein the vehicle data transmission circuit is further structured to select the transmission interval in response to an interval responsive to a priority of the at least a portion of the collected vehicle data** ("creates an event window of a predetermined time interval about the time stamp, and selects a portion of the logged data falling within the predetermined time interval" (para 0017) and "at time interval T2 (e.g., every 60 seconds). Alternately, this initial message may automatically, under certain specified conditions, be transmitted for one or more selected vehicles at a high data rate T1 (e.g., every 200 milliseconds). For example, when the vehicle is selected by a user at the backend, or when a fault or failure condition is recognized for that vehicle, the RDU 20 may automatically begin transmitting at the high rate T1." (para 0099)).

Regarding claim **6**, Mayer discloses and Kim teaches **the apparatus of claim 4**. Additionally, Mayer discloses **wherein the priority relates to transmission of collected data** ("elected" vehicle 3 is in a higher priority of communication than "non-selected" vehicles 2" (para 0058))**.**

Claims 5, 7-16, and 22 are rejected under 35 U.S.C. 103 as being unpatentable over U.S. Pub. No. 2011/0130916 (hereinafter, "Mayer"; previously of record) in view of U.S. Pub. No. 2016/0171793 (hereinafter, "Kim"; newly of record) as applied to claim 1 and 4 above, and in further view of U.S. Pub. No. 2019/0166054 (hereinafter, "Grant"; previously of record).

Regarding claim **5**, Mayer discloses and Kim teaches **the apparatus of claim 4**. However, Mayer does not explicitly teach **wherein the priority relates to a requesting entity.**

Grant, in the same field of endeavor, teaches

wherein the priority relates to a requesting entity (" the data congestion control system 400 can be operable to monitor the resource consumption such as memory and/or CPU of devices 410, 412, 414, 415, 420, and/or 430, to monitor the data priority of the data types being transferred to respective devices 415, 420, and/or 430, and/or to monitor the operating bounds of devices 410, 412, 414, 415, 420, and/or 430, such as the minimum and/or maximum data rate at which these devices can receive and/or transmit data. The data congestion control system can be further operable to calculate a negotiation score of the devices 410, 412, 414, 415, and/or 430" (para 0020)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Grant in order to to calculate a negotiation score of the devices; see Grant at least at [0022].

Regarding claim **7**, Mayer discloses **the apparatus of claim 4**. However, Mayer does not explicitly teach **wherein the priority relates to at least one of:**

a data type; an associated flow; an associated application; an associated vehicle function; a requesting end point; a providing end point; a providing entity; processing of collected data; or storage of collected data.

Grant, in the same field of endeavor, teaches

wherein the priority relates to at least one of:

a data type; an associated flow; an associated application; an associated vehicle function; a requesting end point; a providing end point; a providing entity; processing of collected data; or storage of collected data ("a score and/or ranking of the types of data and/or respective sensor devices" (para 0022))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Grant in order to score and/or rank types of data; see Grant at least at [0022].

Regarding claim **8**, Mayer discloses and Kim teaches **the apparatus of claim 1**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data based at least in part on an amount of the collected vehicle data.**

Grant, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data based at least in part on an amount of the collected vehicle data ("only transmit its collected data when the collected data compares favorably to a threshold" (para 0020) and "the control device 415 may not know the boundary transmission rates of the devices from which data is received" (para 0028))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Grant in order to generate control data for transmission; see Grant at least at [0017].

Regarding claim **9**, Mayer discloses and Kim teaches **the apparatus of claim 8**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data in response to when the amount of the collected vehicle data exceeds a threshold.**

Grant, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data in response to when the amount of the collected vehicle data exceeds a threshold ("The data congestion control system 400 is operable to detect potentially overloaded devices that are receiving data from other devices in the system, and is further operable to initiate a negotiation process reducing the data transmission from some or all of the corresponding devices in the system over time" (para 0020) and "the congestion threshold is based on a maximum transmission rate corresponding to transmission of the aggregated sensor data to the gateway device" (para 0046)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Grant in order to detect potentially overloaded devices; see Grant at least at [0020].

Regarding claim **10**, Mayer discloses and Kim teaches **the apparatus of claim 1**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data by selecting a bandwidth utilization for the at least a portion of the collected vehicle data.**

Grant, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data by selecting a bandwidth utilization for the at least a portion of the collected vehicle data ("cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts" (para 0066)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Grant in order to optimize resource use; see Grant at least at [0066].

Regarding claim **11**, Mayer discloses and Kim teaches **the apparatus of claim 10**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to select the bandwidth utilization in response to at least one of:**

an interval responsive to a historical transmission availability for the vehicle; an operating condition of the vehicle; a bandwidth utilization provided in the vehicle policy data value; a bandwidth utilization responsive to a priority of the at least a portion of the collected vehicle data; or a bandwidth utilization responsive to an availability description for transmitting resources for the at least a portion of the collected vehicle data.

Grant, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to select the bandwidth utilization in response to at least one of:

an interval responsive to a historical transmission availability for the vehicle; an operating condition of the vehicle; a bandwidth utilization provided in the vehicle policy data value; a bandwidth utilization responsive to a priority of the at least a portion of the collected vehicle data; or a bandwidth utilization responsive to an availability description for transmitting resources for the at least a portion of the collected vehicle data ("cloud systems automatically control and optimize

resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts" (para 0066) and "The sensor devices 410, 412, and/or 414 can be constrained devices with small memories, low bandwidth and limited power resources" (para 0018)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Grant in order to optimize resource use; see Grant at least at [0066].

Regarding claim **12**, Mayer discloses and Kim teaches **the apparatus of claim 1**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data by selecting a transmission interval in response to a data type of the at least a portion of the collected vehicle data.**

Grant, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data by selecting a transmission interval in response to a data type of the at least a portion of the collected vehicle data ("monitor the data priority of the data types being transferred to respective devices 415, 420, and/or 430, and/or to monitor the operating bounds of devices 410, 412, 414, 415, 420, and/or 430, such as the minimum and/or maximum data rate at which these devices can receive and/or transmit data" (para 0020) and "determine to reduce the data rate of the different data types by different amounts, for example, where some data types are transmitted at the same or increased rate and/or where other data types are not transmitted at all" (para 0030)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Grant in order to optimize resource use; see Grant at least at [0066].

Regarding claim **13**, Mayer discloses and Kim teaches **the apparatus of claim 1**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data in response to a vehicle operational impact of transmission operations.**

Grant, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data in response to a vehicle operational impact of transmission operations ("This can include a data transmission reduction at the lowest hierarchical levels, where the transmission rate of at least one sensor devices is reduced. Alternatively or in addition, this can include a data transmission reduction at higher levels, where the transmission rate of at least one control device 415 and/or gateway device 430 is reduced. The negotiation process can be performed such that quality of service metrics of each device are honored while enabling each device to operate at within its normal operating boundaries" (para 0020)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Grant in order to enable each device to operate within its normal operating boundaries; see Grant at least at [0020].

Regarding claim **14**, Mayer discloses and Kim teaches **the apparatus of claim 1**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data in response to a power utilization impact of transmission operations.**

Grant, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data in response to a power utilization impact of transmission operations (" instructing a sensor device to hold off in transmitting its data and not

transmit continuously can save battery power while ensuring that data is transmitted when necessary" (para 0033)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Grant in order to save battery power while ensuring that data is transmitted when necessary; see Grant at least at [0033].

Regarding claim **15**, Mayer discloses and Kim teaches **the apparatus of claim 1**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data in response to a data transmission capacity value.**

Grant, in the same field of endeavor, teaches **wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data in response to a data transmission capacity value** ("A control device 415 can determine to begin the congestion negotiation process in response to determining that memory capacity compares unfavorably to a threshold and/or that the incoming transmission rate compares unfavorably to a maximum reception rate threshold" (para 0023)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Grant in order to enable each device to operate within its normal operating boundaries; see Grant at least at [0020].

Regarding claim **16**, Mayer discloses and Kim teaches **the apparatus of claim 15**. Additionally, Mayer discloses **wherein the data transmission capacity value comprises at least one data transmission capacity value selected from the values consisting of:**

a data transmission capacity associated with an access point name; a data transmission capacity associated with an application of the vehicle related to the at least a portion of the collected vehicle data; or a data transmission capacity associated with a vehicle function related to the at least

a portion of the collected vehicle data ; a data transmission capacity associated with a time interval ("the RDU central processor 24 is configured transmit one or more predetermined status messages or vehicle data to the backend or server side of the system at predetermined intervals, with each message containing selected vehicle data or status items." (para 0082) and "GPS data may be down sampled and recorded at periodic intervals of e.g., 500 ms, 1 s, 2 s, 4 s, etc., depending on the desired size of the recorded data log" (para 0190))**;**

a data transmission capacity associated with an entity related to the at least a portion of the collected vehicle data ("the RDS server 30 sends partial status information messages M2 received at a slow data rate T2 from all currently active or online vehicles to the user devices 40 authorized to receive information on the respective vehicles (S-95)" (para 0136))**;**

a data transmission capacity associated with a flow related to the at least a portion of the collected vehicle data ("both selected and non-selected vehicles 3, 2 may transmit the same messages at different rates or different messages at the same rates." (para 0084))**.**

Regarding claim **22**, Mayer discloses and Kim teaches **the apparatus of claim 1**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to adjust the selectively transmitting the at least a portion of the collected vehicle data in response to a quality of service parameter for transmitting operations.**

Grant, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to adjust the selectively transmitting the at least a portion of the collected vehicle data in response to a quality of service parameter for transmitting operations ("The negotiation process can be performed such that quality of service metrics of each device are honored while enabling each device to operate at within its normal operating boundaries" (para 0020))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Grant in order that quality of service metrics of each device are honored; see Grant at least at [0020].

Claims 17-21 are rejected under 35 U.S.C. 103 as being unpatentable over U.S. Pub. No. 2011/0130916 (hereinafter, "Mayer"; previously of record) in view of U.S. Pub. No. 2016/0171793 (hereinafter,"Kim"; newly of record) as applied to claim 1 above, and in further view of U.S. Pub. No. 2018/0261020 (hereinafter, "Petousis"; previously of record).

Regarding claim **17**, Mayer discloses and Kim teaches **the apparatus of claim 1**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data in response to a currently available transmission type.**

Petousis, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data in response to a currently available transmission type (" allocates (e.g., distributes, transfers, optimally forwards) the messages (e.g., vehicle sensor data) to each available communication channel (e.g., based on an optimization analysis)" (para 0045) and " determining available (or anticipated available) network quality, and determining available transformation module and/or transmission module capacity based on the determined network quality." (para 0052))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Petousis in order to have preferential real-time transmission, precluding low-value data from consuming limited communication resources; see Petousis at least at [0016].

Regarding claim **18**, Mayer discloses and Kim teaches **the apparatus of claim 1**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data by selecting a data transmission chunk size for the at least a portion of the collected vehicle data.**

Petousis, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to selectively transmit the at least a portion of the collected vehicle data by selecting a data transmission chunk size for the at least a portion of the collected vehicle data ("the transmission channels are determined according to the size of the vehicle sensor data (e.g., data of large size is designated for high bandwidth channels" (para 0057) and "The data is allocated to the appropriate channel according to size (e.g., large blocks of data are transmitted via the high bandwidth channel) and transmitted" (para 0070))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Petousis in order to have preferential real-time transmission, precluding low-value data from consuming limited communication resources; see Petousis at least at [0016].

Regarding claim **19**, Mayer discloses and Kim teaches **the apparatus of claim 18**. However, Mayer does not explicitly teach **wherein the data transmission chunk size comprises at least one of an individual message size or a single transmission flow size.**

Petousis, in the same field of endeavor, teaches

wherein the data transmission chunk size comprises at least one of an individual message size or a single transmission flow size ("The message header can include the message importance (e.g., determined by prioritizing the vehicle sensor data, below), the message size (e.g., actual or estimated, post-compression, compressibility, etc" (para 0040)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Petousis in order to have preferential real-time transmission, precluding low-value data from consuming limited communication resources; see Petousis at least at [0016].

Regarding claim **20**, Mayer discloses and Kim teaches **the apparatus of claim 18.** However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to select the data transmission chunk size in response to at least one of:**

a data transmission chunk size provided in the vehicle policy data value; a data transmission chunk size responsive to a priority of the at least a portion of the collected vehicle data; a data transmission chunk size responsive to an availability description for transmitting resources for the at least a portion of the collected vehicle data; a data transmission chunk size responsive to a historical transmission availability for the vehicle; or an operating condition of the vehicle.

Petousis, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to select the data transmission chunk size in response to at least one of:

a data transmission chunk size provided in the vehicle policy data value; a data transmission chunk size responsive to a priority of the at least a portion of the collected vehicle data (“ Prioritizing the vehicle sensor data can be based on characteristics of the vehicle sensor data itself (e.g., block size, packet content, compressibility, type, etc.), a prioritization request (e.g., a remote query specifies a type of data or combination of data)”) (para 0028)**;**

a data transmission chunk size responsive to a historical transmission availability for the vehicle; or

an operating condition of the vehicle; a data transmission chunk size responsive to an availability description for transmitting resources for the at least a portion of the collected vehicle

data ("determining a first threshold data size for a first available channel based on available bandwidth" (para 0064)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Petousis in order to have preferential real-time transmission, precluding low-value data from consuming limited communication resources; see Petousis at least at [0016].

Regarding claim **21**, Mayer discloses and Kim teaches **the apparatus of claim 1**. However, Mayer does not explicitly teach **wherein the vehicle data transmission circuit is further structured to adjust the selectively transmitting the at least a portion of the collected vehicle data in response to a success parameter for transmitting operations.**

Petousis, in the same field of endeavor, teaches

wherein the vehicle data transmission circuit is further structured to adjust the selectively transmitting the at least a portion of the collected vehicle data in response to a success parameter for transmitting operations ("Transmitting message data can optionally include waiting for confirmation of successful transmission" (para 0064)).

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Mayer with the teachings of Petousis in order to confirm successful transmission; see Petousis at least at [0064].

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date

of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ADAM ALHARBI whose telephone number is (313)446-6621. The examiner can normally be reached M-F 10am-6:30pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Abby Flynn can be reached on (571) 272-9855. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/ADAM M ALHARBI/
Primary Examiner, Art Unit 3663